University of Colorado at Colorado Springs
Department of Mechanical and Aerospace Engineering

SPCE 5065

HOMEWORK #3

DUE: Friday July 10, 2026

Instructions:

- Include your references at the end of your homework in AIAA format. https://www.aiaa.org/publications/journals/reference-style-and-format
- Include units in your final answers!
- Write non-mathematical answers in complete sentences.
- Clearly state your assumptions

1. For the current events presentations on Thursday 2 July

   a) Summarize the presentation

   b) Describe something you learned from it

   c) Write one question you have left about the presentation

2. What is meant by "altered vestibular functions" and what are its symptoms?

3. What are approximate caloric intake requirements for astronauts? Describe two unique nutritional requirements driven by the free-fall environment.

4. For a Mars mission, what habitable volume would you recommend for the crew quarters on the

a. Flight there

b. Surface

c. Return flight

Explain your rationale.

5. NASA's Bioastronautics Roadmap provides overall ratings for a human Mars mission as shown in the table below. Details may be found here Human Research Roadmap (nasa.gov).

a. Complete the columns for a Ceres mission.

b. Choose three specific areas and discuss them more thoroughly. How would the risk be different for Ceres? What might be a mitigation strategy for that risk?

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Risk	Mars operational	Mars long term	Ceres operational	Ceres long term
Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders				
Risk of Adverse Health and Performance Effects of Celestial Dust Exposure				
Risk of Adverse Health Effects Due to Host-Microorganism Interactions				
Risk of Adverse Health Event Due to Altered Immune Response				
Risk of Adverse Health Outcomes and Decrements in Performance Due to Medical Conditions that occur in Mission, as well as Long Term Health Outcomes Due to Mission Exposures				
Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		N/A		
Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks				
Risk of Bone Fracture due to Spaceflight-induced Changes to Bone				
Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes				
Risk of Impaired Performance Due to Reduced Muscle Size, Strength and Endurance				
Risk of Ineffective or Toxic Medications During Long-Duration Exploration Spaceflight				
Risk of Injury and Compromised Performance Due to EVA Operations				
Risk of Injury from Dynamic Loads				
Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		N/A		
Risk of Performance Decrement and Crew				

Illness Due to Inadequate Food and Nutrition				
Risk of Performance Decrements and Adverse Health Outcomes Resulting from Sleep Loss, Circadian Desynchronization, and Work Overload				
Risk of Radiation Carcinogenesis				
Risk of Reduced Physical Performance Capabilities Due to Reduced Aerobic Capacity				
Risk of Renal Stone Formation				
Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)				

6. Describe the major breakdowns and disconnects that occurred during the Apollo 13 mission. Which elements of the SHELL model were involved?

7. Propose requirements for an EVA suit for a human mission to the following destinations. Include a discussion of the unique aspects of the destination.

a) Moon

b) Mars

c) Ceres

8. NASA has allocated an additional 1,500 kg of launch mass for improving crew health during a 900-day Mars mission. You may invest the mass in only three of the following:

- radiation shielding
- exercise equipment
- food variety
- medical equipment
- private crew quarters
- water reserves
- artificial gravity demonstration hardware
- additional scientific equipment.

Develop a weighted trade study and justify your allocation. Make sure your answer includes:

- objectives
- evaluation criteria
- weighting factors
- trade matrix
- final recommendation

9. Using the SHELL model, compare two human-spaceflight accidents. Evaluate as applicable:

- common failure modes (in the context of the SHELL model)
- organizational influences
- design deficiencies
- operational decisions
- lessons that remain relevant for Artemis or Mars